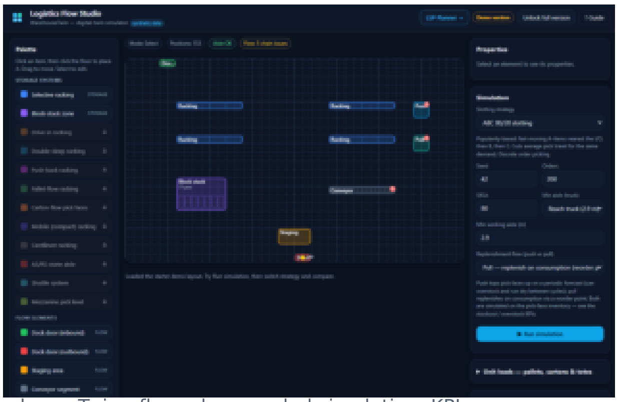
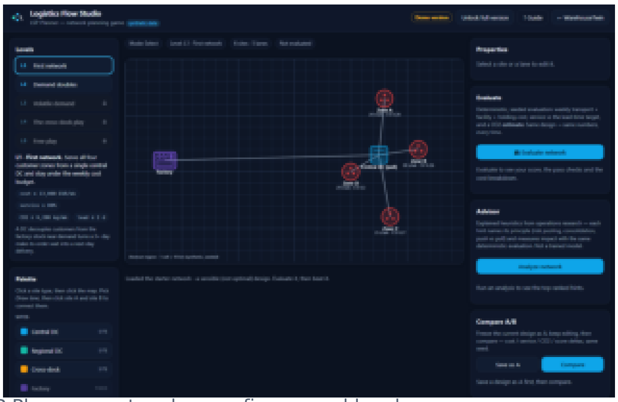
Logistics Flow Studio

one-pager

WarehouseTwin + LSP Planner | offline warehouse and network planning twins, in the browser



WarehouseTwin - floor plan, seeded simulation, KPIs



LSP Planner - network map, five scored levels

WHAT THIS IS

Two small, honest planning games in one repo - a single-page PWA with no build step, no framework, no server and zero runtime network calls. It installs from the browser and runs fully offline.

- WarehouseTwin: draw a warehouse floor (12 storage systems, docks, staging, conveyors), pick slotting and picking strategies, and run a seeded, deterministic simulation with live KPIs, a one-click golden-zone layout optimizer, a pick-travel heatmap and an A/B comparator.

- LSP Planner: place factories, DCs, cross-docks and customer zones; draw FTL or parcel/LTL lanes; toggle push/pull per DC. A pure evaluation engine scores weekly cost, service, a CO_2 estimate and five levels, each built around one network-design lesson.

HEURISTIC ADVISOR - LABELLED AS SUCH

Both apps include an advisor: a transparent rule engine informed by operations-research practice, NOT a trained or black-box model. Every suggestion names its finding, its principle, and an impact measured with the same deterministic simulation.

STANDARDS: ALIGNED-TO, NOT CERTIFIED

Aisle checks are informed by DIN 15185, pallets follow EPAL/EN references, and a reference panel covers ASR A1.8, EN 15512, VDI 2510/3564 and others - as design guidance only. The apps perform no compliance certification of any kind.

USE CASES

- Training: learn slotting, flow design and network economics by playing.

- Pre-study before a WMS or consulting engagement: show which levers matter before commissioning real data work.

- Candidate skill demonstration: domain depth plus engineering discipline (pinned measurements, verification harnesses, CI).

MEASURED RESULTS (SYNTHETIC, SEEDED, TEACHING-SCALE)

Starter demo layout, seed 42, 200 orders, 80 SKUs. Pinned in docs/MEASUREMENTS.md; reproduced headlessly by node test/run-all.mjs.

- ABC 80/20 vs random slotting: 46.71 -> 36.70 m/order pick travel (about -21%).

- One-click layout optimizer: 36.70 -> 18.85 m/order (-48.6%), 5 elements moved, every aisle still valid.

- Heatmap conservation: per-cell walked metres sum exactly to the travel KPI for all five picking strategies.

LSP Planner reference networks (node lsp/verify.js proves each lesson on every run):

- Pull beats push under volatile demand: service 95.0% vs 87.1%, holding 919 vs 1,452 EUR/wk.

- A cross-dock pays off on thin flows: 14,755 vs 16,496 EUR/wk (-10.6%), CO_2 estimate 3,804 vs 5,367 kg/wk.

- Risk pooling vs proximity: a single central DC tops out at 88.4% service and fails the level; adding a regional DC reaches 92.6%.

ENGINEERING DISCIPLINE

- Deterministic by construction: same seed, same numbers, byte-identical.

- Verification harnesses run in CI: optimizer measurement, heatmap conservation invariant, LSP engine gates, offline guard.

- 100% offline: CI greps every app file for external assets and fails on any hit.

DELIVERABLES

- The installable offline PWA itself (WarehouseTwin at /, LSP Planner at /lsp/).

- This one-pager (rebuilt by python tools/make_onepager.py).

- docs/BUSINESS_CASE.md, docs/MEASUREMENTS.md and docs/DOMAIN_NOTES.md - the claims, the numbers, and every model simplification in writing.

Honesty note: all data is synthetic and seeded - no real inventory, orders or telemetry. Figures demonstrate the direction and rough magnitude of well-known levers at teaching scale; they are not forecasts for any real operation.